

## **Abstract of Entrepreneurship Education at Anna University & the courses taught**

1. Developed a 3 credit course on ENGINEERING ENTREPRENEURSHIP DEVELOPMENT for UG B.E.B.TECH. All Disciplines
2. Conceived, Developed, Organised and Delivered the lectures - FDP on Innovation and Entrepreneurship - 1 week Offline 2 weeks Online
3. Conceived, Developed, Organised and Delivered the lectures - AI Product Development Internships - 1 month 45 days - Offline Online
4. Conceived, Developed, Organised and Delivered the lectures - Innovative Product Development Internships - 1 month 45 days - Offline Online
5. Conceived, Developed, Organised and Delivered the lectures - Startup Practice Venture Creation Internships - 1 month 45 days - Offline Online
6. Conceived, Developed, Organised and Delivered the lectures - IoT Product Development Internships - 2 weeks - Offline
7. Conceived, Developed, Organised and Delivered the lectures - Certified Entrepreneurship Educator - 3 Months - Hybrid - as a means of Capacity Building Initiative across the State

## **Entrepreneurship Education at Anna University: A Six-Year Overview**

Over the last six years, Anna University has significantly strengthened its focus on entrepreneurship education, aligning with national priorities such as "Startup India" and "Atmanirbhar Bharat." The university, through its **Centre for Entrepreneurship Development (CED)** and affiliated departments, has introduced and expanded a range of academic and capacity-building courses aimed at fostering innovation, entrepreneurial mindset, and startup readiness among students.

### **1. Core and Elective Courses in Undergraduate (UG) Programs**

Several B.E./B.Tech. programs have integrated entrepreneurship-oriented electives as part of the curriculum:

- **GE8077 – Total Quality Management and Entrepreneurship Development**
  - Offered as an elective to final-year engineering students.
  - Combines principles of TQM with basic entrepreneurial competencies.
- **MG6863 – Engineering Economics and Financial Accounting**
  - Offers economic, financial, and entrepreneurial decision-making concepts.
- **GE6757 – Innovation and Entrepreneurship**
  - Introduced to create a foundational understanding of startup ecosystems, business model canvassing, and ideation processes.
- **OAI1301 – Entrepreneurial Development**
  - An interdisciplinary open elective offered across departments.

- Focuses on entrepreneurial motivation, opportunity identification, and project formulation.

## 2. Postgraduate (PG) Programs

Entrepreneurship and innovation have been incorporated into several PG-level engineering and management programs:

- **MBA in Technology Management and General Management**
  - Offers full courses on entrepreneurship, strategic innovation, and startup financing.
- **Electives such as "Entrepreneurial Finance" and "Managing Startups"**
  - These papers have been designed to cater to students interested in venture creation and startup incubation.

## 3. Value-Added & Certificate Courses

CED Anna University has collaborated with industry and government bodies to offer various short-term and long-term programs:

- **Entrepreneurship Awareness Camps (EACs)** – Funded by NSTEDB, DST, MSME.
- **Faculty Development Programs (FDPs)** on Entrepreneurship – In collaboration with EDII and AICTE.
- **Start-up Bootcamps** in partnership with institutions like Wadhvani Foundation.
- **Design Thinking and Lean Startup Workshops**
  - Offered through E-Cell and Innovation Council units of the university.

## 4. Entrepreneurship Curriculum

As a nodal university, Anna University piloted structured entrepreneurship and innovation-based modules in the current R2023 regulations in the name of UC23E01 ENGINEERING ENTREPRENEURSHIP DEVELOPMENT – 3 credits as compulsory course. These modules include:

- Introduction to Innovation, Design Thinking, and Prototyping.
- Technology Commercialization & Business Validation.
- Pitching, Fundraising & Incubation Process.

## 5. Online & MOOC-based Courses

- Faculty and students have actively taken up **SWAYAM/NPTEL** entrepreneurship courses co-created by Anna University faculty or accessed from leading institutions like IITs and IIMs. Examples include:
  - “Entrepreneurship Essentials”
  - “Business Model Innovation”
  - “Startup India Learning Program”

## Impact and Way Forward

Over the past six years, the structured inclusion of entrepreneurship into mainstream academics has led to increased student startup formations, active innovation clubs, and growing participation in national competitions such as **Smart India Hackathon**, **YUKTI**, **TNSI (Tamilnadu Student Innovator)**, and **STARTIFY**. With continuous updates to syllabi under NEP 2020 and multidisciplinary learning frameworks, Anna University is poised to further integrate entrepreneurial education into all tiers of learning, nurturing job creators over job seekers.

## GE8077 Total Quality Management (Regulation 2017)

- **Subject Code:** GE8077
- **Subject Title:** Total Quality Management
- **Applicability:** Final year undergraduate engineering students across departments
- **Lecture-Tutorial-Practical (L–T–P) & Credits:** 3–0–0, 3 credits

### *Unit I – Introduction (9 hours)*

Need for quality, evolution, definitions, dimensions of quality (product & service) - Basic TQM concepts and framework - Barriers to TQM - Customer focus: orientation, satisfaction, complaints, retention

### *Unit II – TQM Principles (9 hours)*

Leadership and quality statements - Strategic quality planning, quality councils - Employee involvement: motivation, empowerment, teams, recognition & reward, appraisal - Continuous improvement: PDCA (Plan–Do–Check–Act), Supplier partnerships: selection, rating, collaboration

### *Unit III – TQM Tools & Techniques I (9 hours)*

The seven basic quality tools (cause-effect, check sheets, histograms, etc.) - Modern tools: **Six Sigma** (concepts, methodology, application in manufacturing and services including IT) - Benchmarking: rationale and process - FMEA (Failure Mode and Effects Analysis): stages and types

### *Unit IV – TQM Tools & Techniques II (9 hours)*

Quality circles - Cost of Quality - QFD (Quality Function Deployment) - Taguchi quality loss function - TPM (Total Productive Maintenance): concepts and performance measures

### *Unit V – Quality Management Systems (9 hours)*

ISO 9000 series standards: introduction, benefits, and requirements - Sector-specific standards: AS 9100, TS 16949, TL 9000 - System implementation: documentation, internal audits, certification process

## **Objectives & Structure**

The primary aim is to instill an understanding of quality management principles—tools, techniques, systems architecture—and their practical applications across products, services, and global industries. Curriculum is designed as **3 hours of lectures per week**, totaling **45 hours** over the semester (9 hours per Unit)

**OAI1301 – Entrepreneurial Development** course, as per Anna University's curriculum (Regulation 2021 for B.E./B.Tech. – Open Elective):

**Credits:** 3

**Category:** Open Elective (Interdisciplinary)

**Offered by:** Department of Management Studies / Interdisciplinary Elective Cell

### **Course Objectives:**

- To develop entrepreneurial competencies among students.
- To provide insights into the process of identifying and evaluating business opportunities.
- To enable students to prepare business plans and understand the role of institutional support systems for entrepreneurship.
- To create awareness about innovation, startup ecosystem, and funding mechanisms.

### **UNIT I: ENTREPRENEURSHIP FOUNDATIONS**

Concept of Entrepreneurship – Definitions - Entrepreneur – Characteristics, Types  
Intrapreneurship vs. Entrepreneurship - Entrepreneurial motivation – McClelland's theory - Barriers to Entrepreneurship - Role of Entrepreneurs in Economic Development

### **UNIT II: OPPORTUNITY IDENTIFICATION AND PROJECT FORMULATION**

Creativity and Innovation in Entrepreneurship - Opportunity Scanning and Identification  
- Business Ideas – Sources and Selection - Market Feasibility Analysis - Technical and Financial Feasibility - Project Report Preparation and Evaluation

### **UNIT III: BUSINESS PLAN AND START-UP PROCESS**

Business Plan – Meaning, Significance, Components - Lean Canvas and Business Model Canvas - Steps in Starting a Business – Legal Formalities - Intellectual Property Rights (IPR) - Start-up India Policy – DPIIT recognition - Startup Registration and Compliance

### **UNIT IV: ENTREPRENEURIAL SUPPORT SYSTEM**

Institutional Support to Entrepreneurs - MSME, SIDBI, NSIC, EDII, DIC, TIDCO, TANSIDCO, NABARD - Role of Commercial Banks and NBFCs - Incubators, Accelerators, Angel Investors - Venture Capital and Private Equity - Government Schemes – PMEGP, MUDRA, Stand-Up India

## **UNIT V: ENTREPRENEURSHIP IN ACTION**

Women and Social Entrepreneurship - Rural and Agri-Entrepreneurship - Case Studies of Successful Entrepreneurs (Indian and Global) - Sustainable and Green Entrepreneurship - Pitching to Investors – Elevator Pitch, Deck - Emerging Trends – Digital Entrepreneurship, Gig Economy, Circular Economy

### **Course Outcomes:**

Upon completion of this course, students will be able to:

1. Understand the fundamentals of entrepreneurship and the role of entrepreneurs.
2. Identify, evaluate, and select business opportunities.
3. Formulate business plans and understand the startup process.
4. Gain knowledge of institutional support mechanisms for entrepreneurs.
5. Analyze case studies and emerging models in entrepreneurial practice.

### **Textbooks:**

1. Khanka S.S. – *Entrepreneurial Development*, S. Chand Publishing, Latest Edition.
2. Hisrich R.D., Peters M.P. – *Entrepreneurship*, Tata McGraw Hill, Latest Edition.

### **Reference Books:**

- Kuratko, D.F. – *Entrepreneurship: Theory, Process, and Practice*, Cengage Learning.
- Dollinger M.J. – *Entrepreneurship: Strategies and Resources*, Pearson Education.
- Vasant Desai – *Dynamics of Entrepreneurial Development and Management*, Himalaya Publishing.

**GE6757 – Innovation and Entrepreneurship** as per Anna University's regulations (Regulation 2013 and in some cases adopted in later years as an elective):

**Course Code: GE6757**

**Course Title: Innovation and Entrepreneurship**

**Offered to: B.E. / B.Tech. – All branches (as elective or open elective)**

**Semester: Usually 6th or 7th semester**

**Credits: 3**

## **Course Type: Elective**

### **Course Objectives:**

- To create awareness on innovation and entrepreneurship among engineering students.
- To teach the basics of starting and managing a business.
- To develop entrepreneurial competencies including creativity, risk-taking, and leadership.
- To expose students to real-life startup processes and ecosystem elements.

### **Unit I – Introduction to Innovation and Entrepreneurship**

Innovation: Types, characteristics, process of innovation, role in economic development - Entrepreneur: Definition, characteristics, classification - Entrepreneurship: Need, functions, types, myths vs. realities - Distinction between entrepreneur, intrapreneur, technopreneur, and manager

### **Unit II – Entrepreneurial Environment**

Institutional support: DIC, MSME, SIDO, NSIC, SIDBI, EDII, TIIC, etc. - Policies for entrepreneurship in India - Entrepreneurial ecosystem: Role of incubators, accelerators, venture capital - Startup India, Standup India, Atal Innovation Mission (AIM), Make in India

### **Unit III – Business Idea Generation and Feasibility Analysis**

Creativity and innovation in identifying business opportunities - Idea generation techniques – Brainstorming, SCAMPER, TRIZ - Feasibility analysis: Technical, Financial, Market, Legal - Business model canvas and lean startup methodology

### **Unit IV – Business Plan and Project Formulation**

Business plan: Definition, components, importance - Market survey and demand analysis - Project formulation – project report preparation - Break-even analysis, working capital assessment, funding sources

### **Unit V – Entrepreneurial Development and Support**

Entrepreneurial development programs (EDPs): Objectives, phases - Social entrepreneurship and rural entrepreneurship - Women entrepreneurship: Challenges and support mechanisms - Role of E-Cells, IICs, and Innovation Clubs - Intellectual Property Rights (IPR) basics: Patents, Trademarks, Copyrights

### **Text Books:**

1. **S.S. Khanka** – *Entrepreneurial Development*, S. Chand & Co., 2012.
2. **Jayashree Suresh** – *Entrepreneurial Development*, Margham Publications, 2015.

3. **C. B. Gupta & N. P. Srinivasan** – *Entrepreneurship Development*, Sultan Chand & Sons.

#### **Reference Books:**

1. **Hisrich, Peters, Shepherd** – *Entrepreneurship*, 9th Edition, McGraw Hill, 2013.
2. **Bessant and Tidd** – *Innovation and Entrepreneurship*, Wiley India.
3. **Dr. Renu Arora & S.K. Sood** – *Fundamentals of Entrepreneurship and Small Business Management*
4. **Timmons & Spinelli** – *New Venture Creation – Entrepreneurship for the 21st Century*, McGraw Hill.

#### **Course Outcomes:**

At the end of the course, students will be able to:

1. Understand key concepts of innovation and entrepreneurship.
2. Analyze the entrepreneurial environment and identify startup opportunities.
3. Develop and assess a business plan.
4. Apply project formulation and feasibility analysis techniques.
5. Understand the institutional support available for entrepreneurs.

## **MG6863 – Engineering Economics and Financial Accounting**

(Anna University, Regulation 2013), extracted from official and reputable sources:

### **UNIT I: Introduction to Economics (8 hours)**

Flow in an economy - Laws of supply & demand - Engineering vs Economic efficiency - Scope of engineering economics - Cost concepts: marginal cost/revenue, sunk cost, opportunity cost - Break-even analysis (V-ratio) - Elementary economic analysis in: Material selection, Product design, Process planning

### **UNIT II: Value Engineering (10 hours)**

Make-or-buy decisions - Value engineering: functions, aims, procedure - Time value of money & interest formulas: Compound amount factor, Present worth factor, Sinking fund factor, Capital recovery factor, Uniform gradient (annual equivalent) factor, Effective interest rate

### **UNIT III: Cash Flow Analysis (9 hours)**

Comparison bases: Present worth method (revenue- & cost-dominated), Future worth method (revenue- & cost-dominated), Annual equivalent method, Rate of return method

### **UNIT IV: Replacement & Maintenance Analysis (9 hours)**

Maintenance types & replacement scenarios, Determination of an asset's economic life, Replacement analysis: Capital recovery with return, Challenger vs defender concept - Simple probabilistic failure models

#### **UNIT V: Depreciation & Inflation Adjustment (9 hours)**

Depreciation methods: Straight line, Declining balance, Sum-of-years' digits, Sinking fund / annuity method, Service-output method - Evaluation of public alternatives - Inflation-adjusted economic decisions - Examples comparing alternatives & determining economic life

UC23E01 ENGINEERING ENTREPRENEURSHIP DEVELOPMENT and other related contents offered in the recent years is presented in [ANNEXURE 1](#)

<https://drive.google.com/file/d/13LZVnbVPpHBprDpn2chLK5xHQ-gV5hJC/view>